Tameem Bakr

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Research interests: Secure and trustworthy machine learning, watermarking, and collaborative learning.

Education

Mohamed bin Zayed University of Artificial Intelligence — Abu Dhabi, UAE
Doctor of Philosophy in Machine Learning — *Aug. 2026 – Present*
Master of Science in Machine Learning, CGPA: 3.77 — *Aug. 2024 – May 2026*

Egypt-Japan University of Science and Technology — Alexandria, Egypt
Bachelor of Science in Computer Engineering, CGPA: 3.55 — *Sep. 2018 – Mar. 2024*

Publications

- **T. Bakr**, A. Ambreth, and N. Lukas, "Collaborative Threshold Watermarking," in *Proceedings of the 43rd International Conference on Machine Learning (ICML)*, PMLR, vol. 306, pp. 5675–5698, Seoul, South Korea, 2026. [Paper] [arXiv] [Code]
- B. Zhao, L. Kerner, M. Meintz, **T. Bakr**, F. Boenisch, and A. Dziedzic, "Data Provenance for Image Auto-Regressive Generation," in *Proceedings of the Fourteenth International Conference on Learning Representations (ICLR)*, pp. 54973–55009, Rio de Janeiro, Brazil, 2026. [Paper] [arXiv]
- A. Sharshar, Y. Ashraf, **T. Bakr**, S. Hassan, H. Elgendy, M. Yaqub, and M. Guizani, "Not Only Grey Matter: OmniBrain for Robust Multimodal Classification of Alzheimer's Disease," in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV) Workshops*, pp. 1311–1320, 2025. [Paper] [arXiv]
- F. Shamshad, **T. Bakr**, Y. Shaaban, N. Hussein, K. Nandakumar, and N. Lukas, "First-Place Solution to NeurIPS 2024 Invisible Watermark Removal Challenge," Oral presentation at the *ICLR 2025 Workshop on GenAI Watermarking (WMARK)*, Singapore, 2025, and Oral presentation at the *NeurIPS 2025 Workshop on Generative and Protective AI for Content Creation (GenProCC)*, San Diego, USA, 2025. **First place on both tracks.** [WMARK] [GenProCC] [arXiv]
- A. Hussein*, A. Elsetohy*, S. Hadhoud*, **T. Bakr***, Y. Rohaim*, and B. AlKhamissi, "Khattat: Enhancing Readability and Concept Representation of Semantic Typography," in *Computer Vision – ECCV 2024 Workshops, Part V*, Lecture Notes in Computer Science, vol. 15627, pp. 278–295, Springer, 2025. [Paper] [arXiv]
- **T. Bakr**, A. Abdullatif, K. Elzeky, M. Elsayed, and R. Zewail, "Cross-Lingual Transfer Learning for Arabic Signature Verification: Dataset and Baseline Evaluation," in *Applied Machine Learning on Sensing Technologies*, CRC Press, pp. 64–80, 2026. [Paper]
- R. Zewail, **T. Bakr**, and A. Abdullatif, "Resource-Aware Identification of COVID-19 Cough Sounds Using Wavelet Scattering Embeddings," in *2022 2nd International Mobile, Intelligent, and Ubiquitous Computing Conference (MIUCC)*, IEEE, pp. 365–370, Cairo, Egypt, 2022. [Paper]

Under Review and Ongoing Work

- **T. Bakr** and N. Lukas, "ICE: Innovation-based Contribution Estimation," *under review at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR) 2027.*
- **T. Bakr**, Y. Ashraf, and N. Lukas, "Credit and Blame in LLM Multi-Agent Systems: A Survey of Contribution Attribution," *under review at the ACM Conference on Fairness, Accountability, and Transparency (FAccT) 2027.*
- Y. Ashraf, **T. Bakr**, D. S. Chauhan, R. Astudillo, and N. Lukas, "From Traces to Recipes: Constructing and Distilling Reusable Procedural Knowledge into Language Models," *under review at the International Conference on Learning Representations (ICLR) 2027.*
- **T. Bakr** and N. Lukas, "Black-Box Threshold Watermarking for Federated Learning," *in preparation.*
- **T. Bakr** and N. Lukas, "Threshold Watermarking for Language Models," *in preparation.*

*Equal contribution.

Research Experience

Graduate Researcher (Ph.D. and M.Sc.) Aug. 2024 – Present
Mohamed bin Zayed University of Artificial Intelligence *Abu Dhabi, UAE*

- Researching trustworthy machine learning, supervised by *Dr. Nils Lukas*.
- Designing threshold watermarks for federated learning, where any t of n clients can jointly prove model ownership.
- Developing attacks, defenses, and detection methods for content and model watermarking.
- Studying attacks and defenses for machine learning systems, e.g., prompt injection, jailbreaks, and backdoors.
- Investigating how to attribute credit and blame among agents in LLM multi-agent systems.

Research Intern Jul. 2025 – Sep. 2025
CISPA Helmholtz Center for Information Security *Saarland, Germany*

- Conducted research at SprintML Lab, supervised by *Dr. Adam Dziedzic* and *Dr. Franziska Boenisch*.
- Researched watermarking and post-hoc data provenance for image autoregressive models (ICLR 2026).

Machine Learning Research Intern Dec. 2021 – Apr. 2022
Egypt-Japan University of Science and Technology *Alexandria, Egypt*

- Developed resource-efficient COVID-19 detection from cough sounds with wavelet scattering (MIUCC 2022).
- Achieved state-of-the-art performance with lower computational cost.

Industry Experience

AI/ML Engineer, Smartera 3S Solutions and Systems, Alexandria, Egypt Feb. 2024 – May 2024

- Gathered on-site requirements at Ahli Bank Oman for an analytics program of 40 use cases and 100 dashboards.
- Built credit-risk, loan-approval, and churn models with XGBoost, LightGBM, and Bayesian optimization.
- Coordinated data acquisition with the data engineering team and dashboard delivery with the Power BI team.

Software Engineering Intern, GreenMEA, Cairo, Egypt Jan. 2024 – Feb. 2024

- Integrated computer-vision models for PPE-compliance monitoring into a Flask web application.

Data Science Intern, Banque Misr, Cairo, Egypt Aug. 2023 – Sep. 2023

- Optimized ATM cash-replenishment routes using K-means/K-medoids clustering and genetic algorithms.

Honors and Awards

- **Mohamed bin Zayed University of Artificial Intelligence:**
  - Department of Machine Learning Award for High-Impact Publications.
  - Awarded a merit-based, fully funded scholarship to pursue a Ph.D. in Machine Learning.
  - Awarded a merit-based, fully funded scholarship to pursue an M.Sc. in Machine Learning.
- **NeurIPS 2024:** Erasing the Invisible Challenge, first place in both the black-box and beige-box tracks.
- **GITEX 2024:** Cyber Hackathon: Battle of The Elite Capture The Flag Competition (First Place).
- **CyberTalents:** Introduction to Cybersecurity Bootcamp Capture The Flag Competition (First Place).
- **Egypt-Japan University of Science and Technology**: Awarded a merit-based, fully funded scholarship to pursue a B.Sc. in Computer Engineering.

Technical Skills

Watermarking: LLMs (sampling-based, semantic, in-weights), images (post-hoc, in-processing), models (white-box, black-box).
Cryptography: Secret sharing, secure multi-party computation, commitments and zero-knowledge proofs.
Generative Models: LLMs, image autoregressive models, diffusion models.
Deep Learning: PyTorch, torchvision, DeepSpeed, Hugging Face Transformers, Diffusers, PEFT, vLLM.
Scientific Computing: Python, NumPy, SciPy, pandas, scikit-learn, Matplotlib, OpenCV.
Infrastructure: Docker, Git, Linux, SLURM/HPC clusters, Weights & Biases.